Got it—your Chapter 2 must have only **two main sections: 2.1 Related Literature and 2.2 Related Systems**. Here is the revised outline using that format, with ECOSTRESS removed and the annual Landsat 8 design retained.

CHAPTER 2: REVIEW OF RELATED LITERATURE

2.1 Related Literature

This section establishes the concepts, research findings, and methodological foundations of your study.

2.1.1 Land Surface Temperature and Surface Urban Heat Islands

- Definitions and distinctions among LST, atmospheric urban heat islands, and SUHI.
- Environmental factors associated with surface-temperature variation.
- SUHI intensity and the role of a defined non-urban reference.
- Limits of interpreting surface temperature as human heat exposure.

2.1.2 Landsat-Based Environmental Observations

- Landsat 8 surface temperature and surface reflectance products.
- NDVI as a vegetation indicator.
- NDBI as a built-up indicator and its interpretation limits.
- Acquisition timing, cloud screening, and missing observations.
- Annual median composites and comparability across years.
- Spatial resolution, mixed pixels, and the limitations of assigning satellite observations to road locations.

2.1.3 Urban Surface-Temperature Research in Tropical and Philippine Cities

- Findings from Southeast Asian cities.
- Philippine studies linking land cover and surface temperature.
- Existing LST studies in Davao City.
- Urban-area delineation and selection of comparable non-urban reference surfaces.
- Findings and unresolved questions relevant to the present study.

2.1.4 Road Networks and Spatial Graph Representations

- Nodes, edges, adjacency, and connectivity.
- Differences among road connectivity, geographic proximity, and urban geometry.
- OpenStreetMap and OSMnx for representing road networks.
- Intersections, intermediate road points, and node densification.
- Road-adjacency, proximity-based, and other environmental graphs.
- OSM data quality and limitations of historical network representation.

2.1.5 Machine Learning for Annual LST Estimation

- Random Forest regression as a conventional benchmark.
- LSTM for encoding historical environmental sequences.

- Graph neural networks and neighborhood aggregation.
- GraphSAGE for learning node representations.
- Combining temporal encoding and graph aggregation in LSTM–GraphSAGE.
- Relevance of these approaches to the 2019–2024 input sequence and 2025 LST target.

2.1.6 Evaluation of Spatial Prediction and Road-Connectivity Contribution

- Spatial autocorrelation and limitations of random data splitting.
- Shared satellite pixels and information-leakage risks.
- Spatial training, validation, and testing approaches.
- Controlled comparisons of road-connected, proximity-based, and self-only neural configurations.
- Comparison with Random Forest using the same historical predictors and held-out observations.
- RMSE, MAE, and R^2 for LST evaluation.
- Interpretation of derived SUHI maps and limits of predictive conclusions.

2.1.7 Synthesis of Related Literature

- Established findings supporting the selected environmental inputs.
- Evidence supporting investigation of temporal and graph-based models.
- Uncertainty about the added predictive value of road connectivity.
- Research gap addressed by the four-configuration comparison.

2.2 Related Systems

This section reviews **existing implemented models, frameworks, or applications** relevant to your proposed system. Research prototypes can be included; they do not have to be commercial applications. Discuss each system's inputs, processing approach, outputs, evaluation, and limitations.

The headings below are categories for organizing your review. Under each category, identify and discuss actual systems from your selected sources.

2.2.1 Satellite-Based LST and SUHI Mapping Systems

- Systems that process satellite observations to map surface temperature.
- Environmental indicators and reference definitions used.
- Spatial and temporal coverage of their outputs.
- Limitations relevant to annual road-node estimation.

2.2.2 Conventional Machine-Learning Thermal Estimation Systems

- Implemented Random Forest and comparable regression frameworks.
- Input variables and prediction targets.
- Validation approaches and reported performance.
- Treatment of spatial relationships.

2.2.3 Raster-Based Deep-Learning Thermal Mapping Systems

- Implemented CNN or U-Net approaches.
- Representation of spatial context through raster neighborhoods.
- Input data, thermal outputs, and evaluation.
- Relevance and limitations for road-network-based estimation.

2.2.4 Graph-Based Environmental and Thermal Estimation Systems

- Implemented GNN systems for LST, SUHI, or related environmental variables.
- Definitions of nodes, edges, and node features.
- Neighborhood aggregation and temporal-learning approaches.
- Evaluation strategies and reported limitations.
- Differences between their graph representations and the proposed road graph.

2.2.5 Spatiotemporal Temperature Forecasting Systems

- Systems combining historical observations with graph-based spatial relationships.
- Road-surface temperature and atmospheric forecasting applications.
- Differences in observation frequency, prediction horizon, and target variable.
- Components relevant to annual LST estimation and aspects that cannot be directly transferred.

2.2.6 Comparative Analysis of Related Systems

Use a comparison matrix with these columns:

System/author/year | Purpose and target | Data inputs | Temporal scale | Spatial representation | Model | Outputs | Validation | Limitations | Relevance to the proposed system

Compare how the systems handle:

- Historical environmental information.
- Road connectivity or geographic proximity.
- Controls for testing graph contribution.
- Spatial dependence and shared satellite observations.
- LST estimation and SUHI derivation.

2.2.7 Synthesis and Positioning of the Proposed System

- Capabilities and limitations identified across existing systems.
- Justification for the proposed LSTM–GraphSAGE approach.
- Roles of the road-connected, self-only, proximity-based, and Random Forest configurations.
- Intended contribution to annual LST estimation and derived SUHI mapping in Davao City.

Suggested division of work

Member	2.1 Related Literature	2.2 Related Systems
Carla — Thermal foundations, satellite data, and local context	**2.1.1** Land Surface Temperature and SUHI; **2.1.2** Landsat-Based Environmental Observations; **2.1.3** Urban Surface-Temperature Research in Tropical and Philippine Cities	**2.2.1** Satellite-Based LST and SUHI Mapping Systems
Jeff — Road networks, graph representations, and evaluation	**2.1.4** Road Networks and Spatial Graph Representations; **2.1.6** Evaluation of Spatial Prediction and Road-Connectivity Contribution	**2.2.4** Graph-Based Environmental and Thermal Estimation Systems; coordinate **2.2.6** Comparative Analysis of Related Systems
Marce — Prediction models and research synthesis	**2.1.5** Machine Learning for Annual LST Estimation; lead **2.1.7** Synthesis of Related Literature	**2.2.2** Conventional Machine-Learning Thermal Estimation Systems; **2.2.3** Raster-Based Deep-Learning Thermal Mapping Systems; **2.2.5** Spatiotemporal Temperature Forecasting Systems; lead **2.2.7** Synthesis and Positioning of the Proposed System